

$$\begin{array}{ll}
 43. \int x(x-1)^{10} dx & 44. \int x\sqrt{4-x} dx \\
 45. \int (x+1)^2(1-x)^5 dx & 46. \int (x+5)(x-5)^{1/3} dx \\
 47. \int x^3\sqrt{x^2+1} dx & 48. \int 3x^5\sqrt{x^3+1} dx \\
 49. \int \frac{x}{(x^2-4)^3} dx & 50. \int \frac{x}{(x-4)^3} dx
 \end{array}$$

If you do not know what substitution to make, try reducing the integral step by step, using a trial substitution to simplify the integral a bit and then another to simplify it some more. You will see what we mean if you try the sequences of substitutions in Exercises 51 and 52.

$$\begin{array}{l}
 51. \int \frac{18 \tan^2 x \sec^2 x}{(2 + \tan^3 x)^2} dx \\
 \quad \text{a. } u = \tan x, \text{ followed by } v = u^3, \text{ then by } w = 2 + v \\
 \quad \text{b. } u = \tan^3 x, \text{ followed by } v = 2 + u \\
 \quad \text{c. } u = 2 + \tan^3 x \\
 52. \int \sqrt{1 + \sin^2(x-1)} \sin(x-1) \cos(x-1) dx \\
 \quad \text{a. } u = x-1, \text{ followed by } v = \sin u, \text{ then by } w = 1 + v^2 \\
 \quad \text{b. } u = \sin(x-1), \text{ followed by } v = 1 + u^2 \\
 \quad \text{c. } u = 1 + \sin^2(x-1)
 \end{array}$$

Evaluate the integrals in Exercises 53 and 54.

$$\begin{array}{l}
 53. \int \frac{(2r-1) \cos \sqrt{3(2r-1)^2+6}}{\sqrt{3(2r-1)^2+6}} dr \\
 54. \int \frac{\sin \sqrt{\theta}}{\sqrt{\theta} \cos^3 \sqrt{\theta}} d\theta
 \end{array}$$

Initial Value Problems

Solve the initial value problems in Exercises 55–60.

$$\begin{array}{l}
 55. \frac{ds}{dt} = 12t(3t^2-1)^3, \quad s(1) = 3 \\
 56. \frac{dy}{dx} = 4x(x^2+8)^{-1/3}, \quad y(0) = 0 \\
 57. \frac{ds}{dt} = 8 \sin^2\left(t + \frac{\pi}{12}\right), \quad s(0) = 8
 \end{array}$$

$$\begin{array}{l}
 58. \frac{dr}{d\theta} = 3 \cos^2\left(\frac{\pi}{4} - \theta\right), \quad r(0) = \frac{\pi}{8} \\
 59. \frac{d^2s}{dt^2} = -4 \sin\left(2t - \frac{\pi}{2}\right), \quad s'(0) = 100, \quad s(0) = 0 \\
 60. \frac{d^2y}{dx^2} = 4 \sec^2 2x \tan 2x, \quad y'(0) = 4, \quad y(0) = -1
 \end{array}$$

Theory and Examples

61. The velocity of a particle moving back and forth on a line is $v = ds/dt = 6 \sin 2t$ m/sec for all t . If $s = 0$ when $t = 0$, find the value of s when $t = \pi/2$ sec.
62. The acceleration of a particle moving back and forth on a line is $a = d^2s/dt^2 = \pi^2 \cos \pi t$ m/sec² for all t . If $s = 0$ and $v = 8$ m/sec when $t = 0$, find s when $t = 1$ sec.
63. It looks as if we can integrate $2 \sin x \cos x$ with respect to x in three different ways:

$$\begin{array}{ll}
 \text{a. } \int 2 \sin x \cos x dx = \int 2u du & u = \sin x \\
 & = u^2 + C_1 = \sin^2 x + C_1 \\
 \text{b. } \int 2 \sin x \cos x dx = \int -2u du & u = \cos x \\
 & = -u^2 + C_2 = -\cos^2 x + C_2 \\
 \text{c. } \int 2 \sin x \cos x dx = \int \sin 2x dx & 2 \sin x \cos x = \sin 2x \\
 & = -\frac{\cos 2x}{2} + C_3.
 \end{array}$$

Can all three integrations be correct? Give reasons for your answer.

64. (Continuation of Example 9.)

- a. Show by evaluating the integral in the expression

$$\frac{1}{(1/60) - 0} \int_0^{1/60} V_{\max} \sin 120 \pi t dt$$

that the average value of $V = V_{\max} \sin 120 \pi t$ over a full cycle is zero.

- b. The circuit that runs your electric stove is rated 240 volts rms. What is the peak value of the allowable voltage?
- c. Show that

$$\int_0^{1/60} (V_{\max})^2 \sin^2 120 \pi t dt = \frac{(V_{\max})^2}{120}.$$

5.6

Substitution and Area Between Curves

There are two methods for evaluating a definite integral by substitution. One method is to find an antiderivative using substitution and then to evaluate the definite integral by applying the Evaluation Theorem. The other method extends the process of substitution directly to *definite* integrals by changing the limits of integration. We apply the new formula introduced here to the problem of computing the area between two curves.

The Substitution Formula

The following formula shows how the limits of integration change when the variable of integration is changed by substitution.

THEOREM 7—Substitution in Definite Integrals If g' is continuous on the interval $[a, b]$ and f is continuous on the range of $g(x) = u$, then

$$\int_a^b f(g(x)) \cdot g'(x) \, dx = \int_{g(a)}^{g(b)} f(u) \, du.$$

Proof Let F denote any antiderivative of f . Then,

$$\begin{aligned} \int_a^b f(g(x)) \cdot g'(x) \, dx &= F(g(x)) \Big|_{x=a}^{x=b} && \frac{d}{dx} F(g(x)) \\ &= F(g(b)) - F(g(a)) && = F'(g(x))g'(x) \\ &= F(u) \Big|_{u=g(a)}^{u=g(b)} && = f(g(x))g'(x) \\ &= \int_{g(a)}^{g(b)} f(u) \, du. && \text{Fundamental Theorem, Part 2} \quad \blacksquare \end{aligned}$$

To use the formula, make the same u -substitution $u = g(x)$ and $du = g'(x) \, dx$ you would use to evaluate the corresponding indefinite integral. Then integrate the transformed integral with respect to u from the value $g(a)$ (the value of u at $x = a$) to the value $g(b)$ (the value of u at $x = b$).

EXAMPLE 1 Evaluate $\int_{-1}^1 3x^2 \sqrt{x^3 + 1} \, dx$.

Solution We have two choices.

Method 1: Transform the integral and evaluate the transformed integral with the transformed limits given in Theorem 7.

$$\begin{aligned} \int_{-1}^1 3x^2 \sqrt{x^3 + 1} \, dx &&& \begin{aligned} \text{Let } u &= x^3 + 1, du = 3x^2 \, dx. \\ \text{When } x &= -1, u = (-1)^3 + 1 = 0. \\ \text{When } x &= 1, u = (1)^3 + 1 = 2. \end{aligned} \\ &= \int_0^2 \sqrt{u} \, du \\ &= \frac{2}{3} u^{3/2} \Big|_0^2 && \text{Evaluate the new definite integral.} \\ &= \frac{2}{3} [2^{3/2} - 0^{3/2}] = \frac{2}{3} [2\sqrt{2}] = \frac{4\sqrt{2}}{3} \end{aligned}$$

Method 2: Transform the integral as an indefinite integral, integrate, change back to x , and use the original x -limits.

$$\begin{aligned} \int 3x^2 \sqrt{x^3 + 1} \, dx &= \int \sqrt{u} \, du && \text{Let } u = x^3 + 1, du = 3x^2 \, dx. \\ &= \frac{2}{3} u^{3/2} + C && \text{Integrate with respect to } u. \\ &= \frac{2}{3} (x^3 + 1)^{3/2} + C && \text{Replace } u \text{ by } x^3 + 1. \\ \int_{-1}^1 3x^2 \sqrt{x^3 + 1} \, dx &= \frac{2}{3} (x^3 + 1)^{3/2} \Big|_{-1}^1 && \text{Use the integral just found, with} \\ &= \frac{2}{3} [(1)^3 + 1)^{3/2} - ((-1)^3 + 1)^{3/2}] && \text{limits of integration for } x. \\ &= \frac{2}{3} [2^{3/2} - 0^{3/2}] = \frac{2}{3} [2\sqrt{2}] = \frac{4\sqrt{2}}{3} \quad \blacksquare \end{aligned}$$

Which method is better—evaluating the transformed definite integral with transformed limits using Theorem 7, or transforming the integral, integrating, and transforming back to use the original limits of integration? In Example 1, the first method seems easier, but that is not always the case. Generally, it is best to know both methods and to use whichever one seems better at the time.

EXAMPLE 2 We use the method of transforming the limits of integration.

$$\begin{aligned}
 \int_{\pi/4}^{\pi/2} \cot \theta \csc^2 \theta \, d\theta &= \int_1^0 u \cdot (-du) \\
 &= -\int_1^0 u \, du \\
 &= -\left[\frac{u^2}{2}\right]_1^0 \\
 &= -\left[\frac{(0)^2}{2} - \frac{(1)^2}{2}\right] = \frac{1}{2}
 \end{aligned}$$

Let $u = \cot \theta$, $du = -\csc^2 \theta \, d\theta$,
 $-du = \csc^2 \theta \, d\theta$.
 When $\theta = \pi/4$, $u = \cot(\pi/4) = 1$.
 When $\theta = \pi/2$, $u = \cot(\pi/2) = 0$.

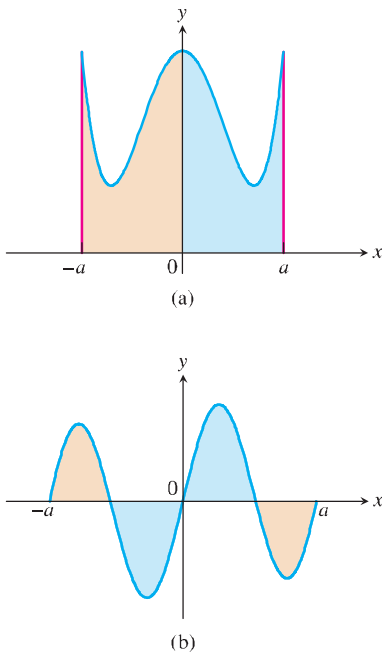


FIGURE 5.24 (a) f even, $\int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx$
 (b) f odd, $\int_{-a}^a f(x) \, dx = 0$

Definite Integrals of Symmetric Functions

The Substitution Formula in Theorem 7 simplifies the calculation of definite integrals of even and odd functions (Section 1.1) over a symmetric interval $[-a, a]$ (Figure 5.24).

THEOREM 8 Let f be continuous on the symmetric interval $[-a, a]$.

- (a) If f is even, then $\int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx$.
 (b) If f is odd, then $\int_{-a}^a f(x) \, dx = 0$.

Proof of Part (a)

$$\begin{aligned}
 \int_{-a}^a f(x) \, dx &= \int_{-a}^0 f(x) \, dx + \int_0^a f(x) \, dx && \text{Additivity Rule for Definite Integrals} \\
 &= -\int_0^{-a} f(x) \, dx + \int_0^a f(x) \, dx && \text{Order of Integration Rule} \\
 &= -\int_0^a f(-u)(-du) + \int_0^a f(x) \, dx && \text{Let } u = -x, \, du = -dx. \\
 &&& \text{When } x = 0, \, u = 0. \\
 &&& \text{When } x = -a, \, u = a. \\
 &= \int_0^a f(-u) \, du + \int_0^a f(x) \, dx \\
 &= \int_0^a f(u) \, du + \int_0^a f(x) \, dx && f \text{ is even, so } f(-u) = f(u). \\
 &= 2 \int_0^a f(x) \, dx
 \end{aligned}$$

The proof of part (b) is entirely similar and you are asked to give it in Exercise 86. ■

The assertions of Theorem 8 remain true when f is an integrable function (rather than having the stronger property of being continuous).

EXAMPLE 3 Evaluate $\int_{-2}^2 (x^4 - 4x^2 + 6) dx$.

Solution Since $f(x) = x^4 - 4x^2 + 6$ satisfies $f(-x) = f(x)$, it is even on the symmetric interval $[-2, 2]$, so

$$\begin{aligned} \int_{-2}^2 (x^4 - 4x^2 + 6) dx &= 2 \int_0^2 (x^4 - 4x^2 + 6) dx \\ &= 2 \left[\frac{x^5}{5} - \frac{4}{3}x^3 + 6x \right]_0^2 \\ &= 2 \left(\frac{32}{5} - \frac{32}{3} + 12 \right) = \frac{232}{15}. \end{aligned}$$

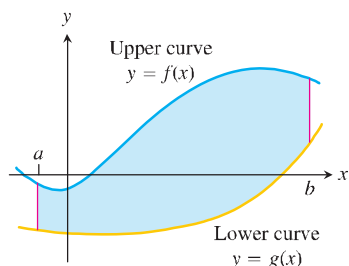


FIGURE 5.25 The region between the curves $y = f(x)$ and $y = g(x)$ and the lines $x = a$ and $x = b$.

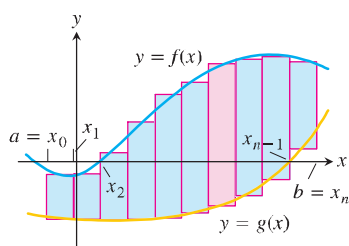


FIGURE 5.26 We approximate the region with rectangles perpendicular to the x -axis.

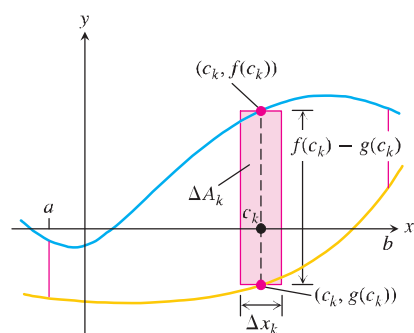


FIGURE 5.27 The area ΔA_k of the k th rectangle is the product of its height, $f(c_k) - g(c_k)$, and its width, Δx_k .

Areas Between Curves

Suppose we want to find the area of a region that is bounded above by the curve $y = f(x)$, below by the curve $y = g(x)$, and on the left and right by the lines $x = a$ and $x = b$ (Figure 5.25). The region might accidentally have a shape whose area we could find with geometry, but if f and g are arbitrary continuous functions, we usually have to find the area with an integral.

To see what the integral should be, we first approximate the region with n vertical rectangles based on a partition $P = \{x_0, x_1, \dots, x_n\}$ of $[a, b]$ (Figure 5.26). The area of the k th rectangle (Figure 5.27) is

$$\Delta A_k = \text{height} \times \text{width} = [f(c_k) - g(c_k)] \Delta x_k.$$

We then approximate the area of the region by adding the areas of the n rectangles:

$$A \approx \sum_{k=1}^n \Delta A_k = \sum_{k=1}^n [f(c_k) - g(c_k)] \Delta x_k. \quad \text{Riemann sum}$$

As $\|P\| \rightarrow 0$, the sums on the right approach the limit $\int_a^b [f(x) - g(x)] dx$ because f and g are continuous. We take the area of the region to be the value of this integral. That is,

$$A = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n [f(c_k) - g(c_k)] \Delta x_k = \int_a^b [f(x) - g(x)] dx.$$

DEFINITION If f and g are continuous with $f(x) \geq g(x)$ throughout $[a, b]$, then the **area of the region between the curves $y = f(x)$ and $y = g(x)$ from a to b** is the integral of $(f - g)$ from a to b :

$$A = \int_a^b [f(x) - g(x)] dx.$$

When applying this definition it is helpful to graph the curves. The graph reveals which curve is the upper curve f and which is the lower curve g . It also helps you find the limits of integration if they are not given. You may need to find where the curves intersect to

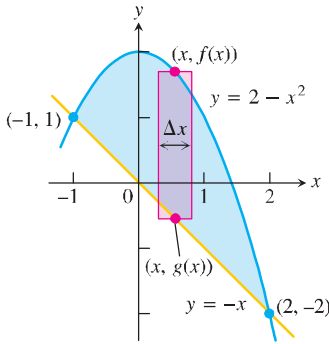


FIGURE 5.28 The region in Example 4 with a typical approximating rectangle.

determine the limits of integration, and this may involve solving the equation $f(x) = g(x)$ for values of x . Then you can integrate the function $f - g$ for the area between the intersections.

EXAMPLE 4 Find the area of the region enclosed by the parabola $y = 2 - x^2$ and the line $y = -x$.

Solution First we sketch the two curves (Figure 5.28). The limits of integration are found by solving $y = 2 - x^2$ and $y = -x$ simultaneously for x .

$$\begin{aligned}
 2 - x^2 &= -x && \text{Equate } f(x) \text{ and } g(x). \\
 x^2 - x - 2 &= 0 && \text{Rewrite.} \\
 (x + 1)(x - 2) &= 0 && \text{Factor.} \\
 x = -1, \quad x = 2. &&& \text{Solve.}
 \end{aligned}$$

The region runs from $x = -1$ to $x = 2$. The limits of integration are $a = -1$, $b = 2$. The area between the curves is

$$\begin{aligned}
 A &= \int_a^b [f(x) - g(x)] \, dx = \int_{-1}^2 [(2 - x^2) - (-x)] \, dx \\
 &= \int_{-1}^2 (2 + x - x^2) \, dx = \left[2x + \frac{x^2}{2} - \frac{x^3}{3} \right]_{-1}^2 \\
 &= \left(4 + \frac{4}{2} - \frac{8}{3} \right) - \left(-2 + \frac{1}{2} + \frac{1}{3} \right) = \frac{9}{2}
 \end{aligned}$$

HISTORICAL BIOGRAPHY

Richard Dedekind
(1831–1916)

If the formula for a bounding curve changes at one or more points, we subdivide the region into subregions that correspond to the formula changes and apply the formula for the area between curves to each subregion.

EXAMPLE 5 Find the area of the region in the first quadrant that is bounded above by $y = \sqrt{x}$ and below by the x -axis and the line $y = x - 2$.

Solution The sketch (Figure 5.29) shows that the region's upper boundary is the graph of $f(x) = \sqrt{x}$. The lower boundary changes from $g(x) = 0$ for $0 \leq x \leq 2$ to $g(x) = x - 2$ for $2 \leq x \leq 4$ (both formulas agree at $x = 2$). We subdivide the region at $x = 2$ into subregions A and B , shown in Figure 5.29.

The limits of integration for region A are $a = 0$ and $b = 2$. The left-hand limit for region B is $a = 2$. To find the right-hand limit, we solve the equations $y = \sqrt{x}$ and $y = x - 2$ simultaneously for x :

$$\begin{aligned}
 \sqrt{x} &= x - 2 && \text{Equate } f(x) \text{ and } g(x). \\
 x &= (x - 2)^2 = x^2 - 4x + 4 && \text{Square both sides.} \\
 x^2 - 5x + 4 &= 0 && \text{Rewrite.} \\
 (x - 1)(x - 4) &= 0 && \text{Factor.} \\
 x = 1, \quad x = 4. &&& \text{Solve.}
 \end{aligned}$$

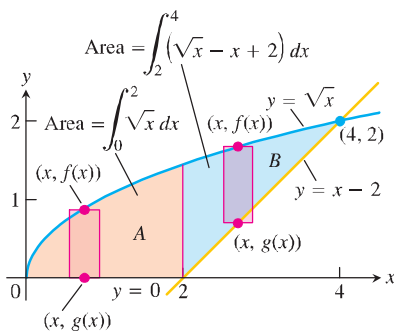


FIGURE 5.29 When the formula for a bounding curve changes, the area integral changes to become the sum of integrals to match, one integral for each of the shaded regions shown here for Example 5.

Only the value $x = 4$ satisfies the equation $\sqrt{x} = x - 2$. The value $x = 1$ is an extraneous root introduced by squaring. The right-hand limit is $b = 4$.

$$\begin{aligned}
 \text{For } 0 \leq x \leq 2: \quad f(x) - g(x) &= \sqrt{x} - 0 = \sqrt{x} \\
 \text{For } 2 \leq x \leq 4: \quad f(x) - g(x) &= \sqrt{x} - (x - 2) = \sqrt{x} - x + 2
 \end{aligned}$$

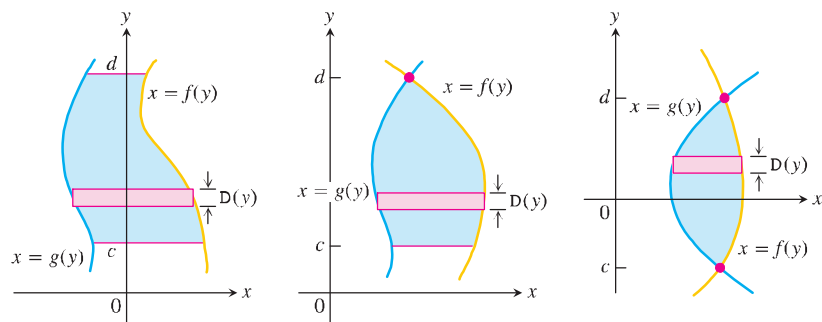
We add the areas of subregions A and B to find the total area:

$$\begin{aligned}
 \text{Total area} &= \underbrace{\int_0^2 \sqrt{x} \, dx}_{\text{area of } A} + \underbrace{\int_2^4 (\sqrt{x} - x + 2) \, dx}_{\text{area of } B} \\
 &= \left[\frac{2}{3} x^{3/2} \right]_0^2 + \left[\frac{2}{3} x^{3/2} - \frac{x^2}{2} + 2x \right]_2^4 \\
 &= \frac{2}{3} (2)^{3/2} - 0 + \left(\frac{2}{3} (4)^{3/2} - 8 + 8 \right) - \left(\frac{2}{3} (2)^{3/2} - 2 + 4 \right) \\
 &= \frac{2}{3} (8) - 2 = \frac{10}{3}.
 \end{aligned}$$

Integration with Respect to y

If a region's bounding curves are described by functions of y , the approximating rectangles are horizontal instead of vertical and the basic formula has y in place of x .

For regions like these:



use the formula

$$A = \int_c^d [f(y) - g(y)] \, dy.$$

In this equation f always denotes the right-hand curve and g the left-hand curve, so $f(y) - g(y)$ is nonnegative.

EXAMPLE 6 Find the area of the region in Example 5 by integrating with respect to y .

Solution We first sketch the region and a typical *horizontal* rectangle based on a partition of an interval of y -values (Figure 5.30). The region's right-hand boundary is the line $x = y + 2$, so $f(y) = y + 2$. The left-hand boundary is the curve $x = y^2$, so $g(y) = y^2$. The lower limit of integration is $y = 0$. We find the upper limit by solving $x = y + 2$ and $x = y^2$ simultaneously for y :

$$\begin{aligned}
 y + 2 &= y^2 && \text{Equate } f(y) = y + 2 \text{ and } g(y) = y^2. \\
 y^2 - y - 2 &= 0 && \text{Rewrite.} \\
 (y + 1)(y - 2) &= 0 && \text{Factor.} \\
 y = -1, \quad y = 2 &&& \text{Solve.}
 \end{aligned}$$

The upper limit of integration is $b = 2$. (The value $y = -1$ gives a point of intersection *below* the x -axis.)

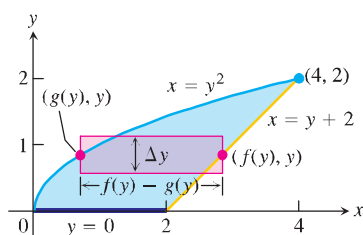


FIGURE 5.30 It takes two integrations to find the area of this region if we integrate with respect to x . It takes only one if we integrate with respect to y (Example 6).

The area of the region is

$$\begin{aligned}
 A &= \int_c^d [f(y) - g(y)] dy = \int_0^2 [y + 2 - y^2] dy \\
 &= \int_0^2 [2 + y - y^2] dy \\
 &= \left[2y + \frac{y^2}{2} - \frac{y^3}{3} \right]_0^2 \\
 &= 4 + \frac{4}{2} - \frac{8}{3} = \frac{10}{3}.
 \end{aligned}$$

This is the result of Example 5, found with less work. ■

Exercises 5.6

Evaluating Definite Integrals

Use the Substitution Formula in Theorem 7 to evaluate the integrals in Exercises 1–24.

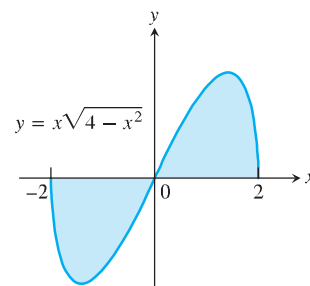
1. a. $\int_0^3 \sqrt{y+1} dy$ b. $\int_{-1}^0 \sqrt{y+1} dy$
2. a. $\int_0^1 r\sqrt{1-r^2} dr$ b. $\int_{-1}^1 r\sqrt{1-r^2} dr$
3. a. $\int_0^{\pi/4} \tan x \sec^2 x dx$ b. $\int_{-\pi/4}^0 \tan x \sec^2 x dx$
4. a. $\int_0^{\pi} 3 \cos^2 x \sin x dx$ b. $\int_{2\pi}^{3\pi} 3 \cos^2 x \sin x dx$
5. a. $\int_0^1 t^3(1+t^4)^3 dt$ b. $\int_{-1}^1 t^3(1+t^4)^3 dt$
6. a. $\int_0^{\sqrt{7}} t(t^2+1)^{1/3} dt$ b. $\int_{-\sqrt{7}}^0 t(t^2+1)^{1/3} dt$
7. a. $\int_{-1}^1 \frac{5r}{(4+r^2)^2} dr$ b. $\int_0^1 \frac{5r}{(4+r^2)^2} dr$
8. a. $\int_0^1 \frac{10\sqrt{v}}{(1+v^{3/2})^2} dv$ b. $\int_1^4 \frac{10\sqrt{v}}{(1+v^{3/2})^2} dv$
9. a. $\int_0^{\sqrt{3}} \frac{4x}{\sqrt{x^2+1}} dx$ b. $\int_{-\sqrt{3}}^{\sqrt{3}} \frac{4x}{\sqrt{x^2+1}} dx$
10. a. $\int_0^1 \frac{x^3}{\sqrt{x^4+9}} dx$ b. $\int_{-1}^0 \frac{x^3}{\sqrt{x^4+9}} dx$
11. a. $\int_0^{\pi/6} (1 - \cos 3t) \sin 3t dt$ b. $\int_{\pi/6}^{\pi/3} (1 - \cos 3t) \sin 3t dt$
12. a. $\int_{-\pi/2}^0 \left(2 + \tan \frac{t}{2}\right) \sec^2 \frac{t}{2} dt$ b. $\int_{-\pi/2}^{\pi/2} \left(2 + \tan \frac{t}{2}\right) \sec^2 \frac{t}{2} dt$
13. a. $\int_0^{2\pi} \frac{\cos z}{\sqrt{4+3\sin z}} dz$ b. $\int_{-\pi}^{\pi} \frac{\cos z}{\sqrt{4+3\sin z}} dz$
14. a. $\int_{-\pi/2}^0 \frac{\sin w}{(3+2\cos w)^2} dw$ b. $\int_0^{\pi/2} \frac{\sin w}{(3+2\cos w)^2} dw$
15. $\int_0^1 \sqrt{t^5+2t}(5t^4+2) dt$ 16. $\int_1^4 \frac{dy}{2\sqrt{y}(1+\sqrt{y})^2}$

17. $\int_0^{\pi/6} \cos^{-3} 2\theta \sin 2\theta d\theta$ 18. $\int_{\pi}^{3\pi/2} \cot^5 \left(\frac{\theta}{6}\right) \sec^2 \left(\frac{\theta}{6}\right) d\theta$
19. $\int_0^{\pi} 5(5-4\cos t)^{1/4} \sin t dt$ 20. $\int_0^{\pi/4} (1 - \sin 2t)^{3/2} \cos 2t dt$
21. $\int_0^1 (4y - y^2 + 4y^3 + 1)^{-2/3} (12y^2 - 2y + 4) dy$
22. $\int_0^1 (y^3 + 6y^2 - 12y + 9)^{-1/2} (y^2 + 4y - 4) dy$
23. $\int_0^{\sqrt[3]{\pi}} \sqrt{\theta} \cos^2(\theta^{3/2}) d\theta$ 24. $\int_{-1}^{-1/2} t^{-2} \sin^2 \left(1 + \frac{1}{t}\right) dt$

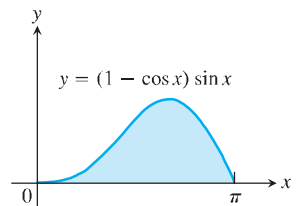
Area

Find the total areas of the shaded regions in Exercises 25–40.

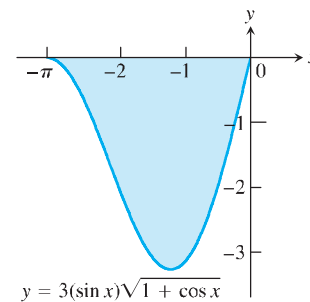
25.



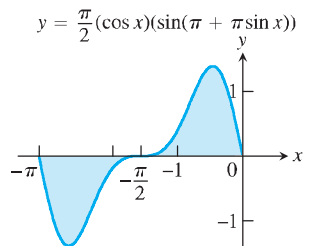
26.



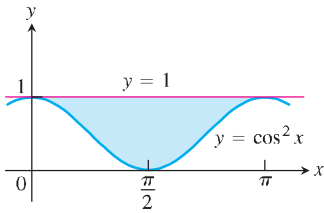
27.



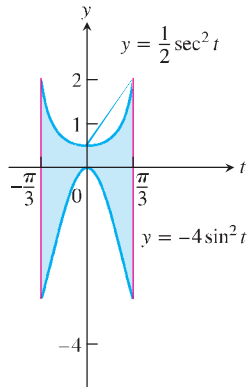
28.



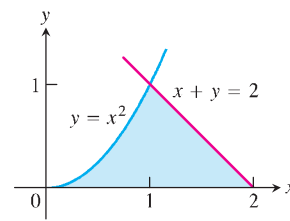
29.



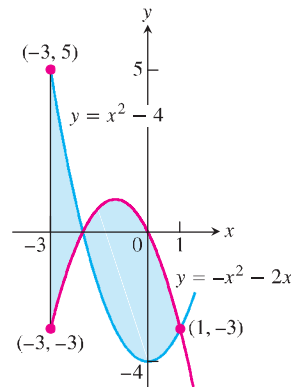
30.



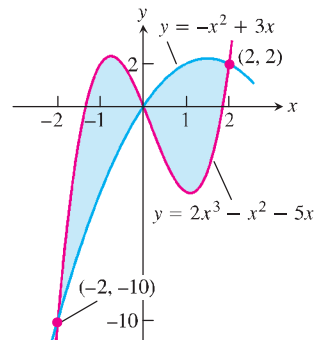
36.



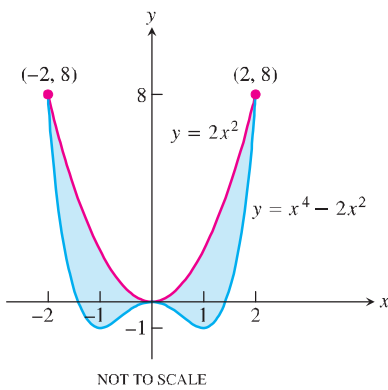
37.



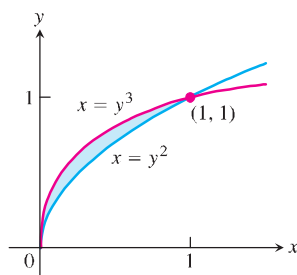
38.



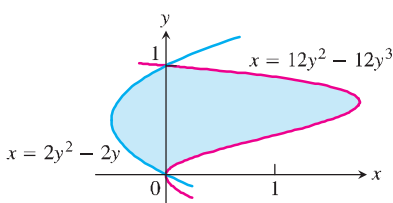
31.



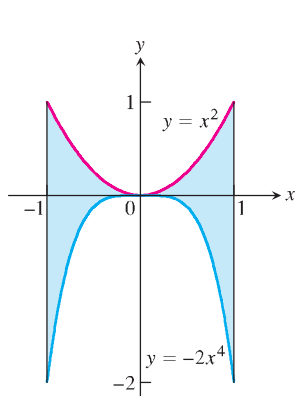
32.



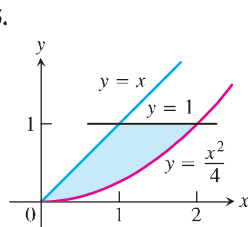
33.



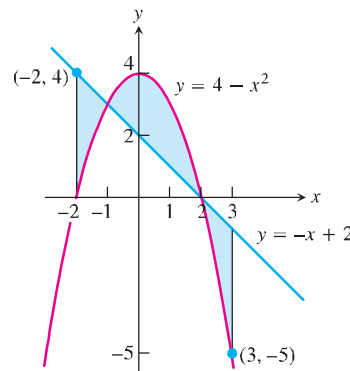
34.



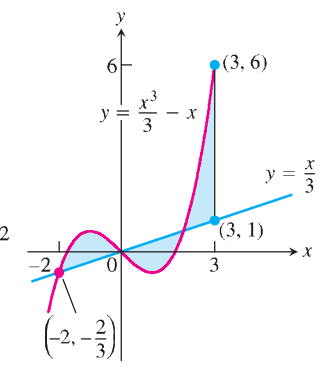
35.



39.



40.



Find the areas of the regions enclosed by the lines and curves in Exercises 41–50.

41. $y = x^2 - 2$ and $y = 2$

42. $y = 2x - x^2$ and $y = -3$

43. $y = x^4$ and $y = 8x$

44. $y = x^2 - 2x$ and $y = x$

45. $y = x^2$ and $y = -x^2 + 4x$

46. $y = 7 - 2x^2$ and $y = x^2 + 4$

47. $y = x^4 - 4x^2 + 4$ and $y = x^2$

48. $y = x\sqrt{a^2 - x^2}$, $a > 0$, and $y = 0$

49. $y = \sqrt{|x|}$ and $5y = x + 6$ (How many intersection points are there?)

50. $y = |x^2 - 4|$ and $y = (x^2/2) + 4$

Find the areas of the regions enclosed by the lines and curves in Exercises 51–58.

51. $x = 2y^2$, $x = 0$, and $y = 3$

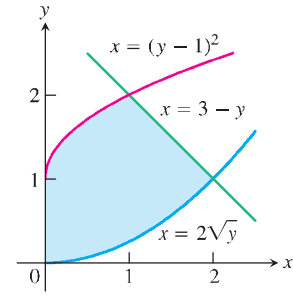
52. $x = y^2$ and $x = y + 2$
 53. $y^2 - 4x = 4$ and $4x - y = 16$
 54. $x - y^2 = 0$ and $x + 2y^2 = 3$
 55. $x = y^2 - y$ and $x = 2y^2 - 2y - 6$
 56. $x - y^{2/3} = 0$ and $x + y^4 = 2$
 57. $x = y^2 - 1$ and $x = |y|\sqrt{1 - y^2}$
 58. $x = y^3 - y^2$ and $x = 2y$

Find the areas of the regions enclosed by the curves in Exercises 59–62.

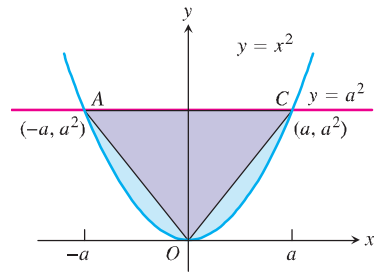
59. $4x^2 + y = 4$ and $x^4 - y = 1$
 60. $x^3 - y = 0$ and $3x^2 - y = 4$
 61. $x + 4y^2 = 4$ and $x + y^4 = 1$, for $x \geq 0$
 62. $x + y^2 = 3$ and $4x + y^2 = 0$

Find the areas of the regions enclosed by the lines and curves in Exercises 63–70.

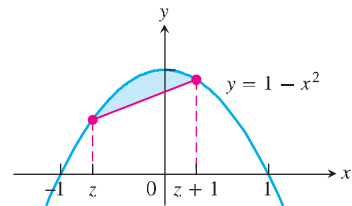
63. $y = 2 \sin x$ and $y = \sin 2x$, $0 \leq x \leq \pi$
 64. $y = 8 \cos x$ and $y = \sec^2 x$, $-\pi/3 \leq x \leq \pi/3$
 65. $y = \cos(\pi x/2)$ and $y = 1 - x^2$
 66. $y = \sin(\pi x/2)$ and $y = x$
 67. $y = \sec^2 x$, $y = \tan^2 x$, $x = -\pi/4$, and $x = \pi/4$
 68. $x = \tan^2 y$ and $x = -\tan^2 y$, $-\pi/4 \leq y \leq \pi/4$
 69. $x = 3 \sin y \sqrt{\cos y}$ and $x = 0$, $0 \leq y \leq \pi/2$
 70. $y = \sec^2(\pi x/3)$ and $y = x^{1/3}$, $-1 \leq x \leq 1$
 71. Find the area of the propeller-shaped region enclosed by the curve $x - y^3 = 0$ and the line $x - y = 0$.
 72. Find the area of the propeller-shaped region enclosed by the curves $x - y^{1/3} = 0$ and $x - y^{1/5} = 0$.
 73. Find the area of the region in the first quadrant bounded by the line $y = x$, the line $x = 2$, the curve $y = 1/x^2$, and the x -axis.
 74. Find the area of the “triangular” region in the first quadrant bounded on the left by the y -axis and on the right by the curves $y = \sin x$ and $y = \cos x$.
 75. The region bounded below by the parabola $y = x^2$ and above by the line $y = 4$ is to be partitioned into two subsections of equal area by cutting across it with the horizontal line $y = c$.
 a. Sketch the region and draw a line $y = c$ across it that looks about right. In terms of c , what are the coordinates of the points where the line and parabola intersect? Add them to your figure.
 b. Find c by integrating with respect to y . (This puts c in the limits of integration.)
 c. Find c by integrating with respect to x . (This puts c into the integrand as well.)
 76. Find the area of the region between the curve $y = 3 - x^2$ and the line $y = -1$ by integrating with respect to (a) x , (b) y .
 77. Find the area of the region in the first quadrant bounded on the left by the y -axis, below by the line $y = x/4$, above left by the curve $y = 1 + \sqrt{x}$, and above right by the curve $y = 2/\sqrt{x}$.
 78. Find the area of the region in the first quadrant bounded on the left by the y -axis, below by the curve $x = 2\sqrt{y}$, above left by the curve $x = (y - 1)^2$, and above right by the line $x = 3 - y$.



79. The figure here shows triangle AOC inscribed in the region cut from the parabola $y = x^2$ by the line $y = a^2$. Find the limit of the ratio of the area of the triangle to the area of the parabolic region as a approaches zero.



80. Suppose the area of the region between the graph of a positive continuous function f and the x -axis from $x = a$ to $x = b$ is 4 square units. Find the area between the curves $y = f(x)$ and $y = 2f(x)$ from $x = a$ to $x = b$.
 81. Show that the area of the shaded region equals $1/6$ for all values of z .



82. True, sometimes true, or never true? The area of the region between the graphs of the continuous functions $y = f(x)$ and $y = g(x)$ and the vertical lines $x = a$ and $x = b$ ($a < b$) is

$$\int_a^b [f(x) - g(x)] dx.$$

Give reasons for your answer.

Theory and Examples

83. Suppose that $F(x)$ is an antiderivative of $f(x) = (\sin x)/x$, $x > 0$. Express

$$\int_1^3 \frac{\sin 2x}{x} dx$$

in terms of F .

84. Show that if
- f
- is continuous, then

$$\int_0^1 f(x) dx = \int_0^1 f(1-x) dx.$$

85. Suppose that

$$\int_0^1 f(x) dx = 3.$$

Find

$$\int_{-1}^0 f(x) dx$$

if (a) f is odd, (b) f is even.

86. a. Show that if
- f
- is odd on
- $[-a, a]$
- , then

$$\int_{-a}^a f(x) dx = 0.$$

b. Test the result in part (a) with $f(x) = \sin x$ and $a = \pi/2$.

87. If
- f
- is a continuous function, find the value of the integral

$$I = \int_0^a \frac{f(x) dx}{f(x) + f(a-x)}$$

by making the substitution $u = a - x$ and adding the resulting integral to I .

88. By using a substitution, prove that for all positive numbers
- x
- and
- y
- ,

$$\int_x^{xy} \frac{1}{t} dt = \int_1^y \frac{1}{t} dt.$$

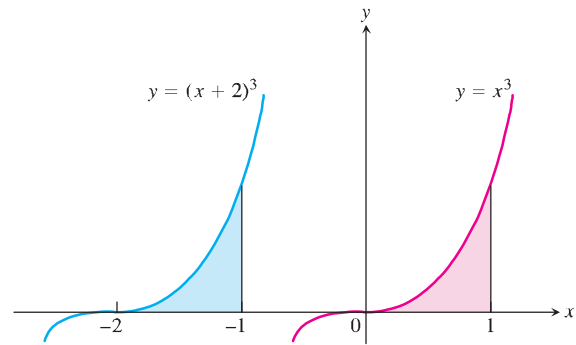
The Shift Property for Definite Integrals A basic property of definite integrals is their invariance under translation, as expressed by the equation

$$\int_a^b f(x) dx = \int_{a-c}^{b-c} f(x+c) dx. \quad (1)$$

The equation holds whenever f is integrable and defined for the necessary values of x . For example, in the accompanying figure show that

$$\int_{-2}^{-1} (x+2)^3 dx = \int_0^1 x^3 dx$$

because the areas of the shaded regions are congruent.



89. Use a substitution to verify Equation (1).

90. For each of the following functions, graph
- $f(x)$
- over
- $[a, b]$
- and
- $f(x+c)$
- over
- $[a-c, b-c]$
- to convince yourself that Equation (1) is reasonable.

a. $f(x) = x^2$, $a = 0$, $b = 1$, $c = 1$

b. $f(x) = \sin x$, $a = 0$, $b = \pi$, $c = \pi/2$

c. $f(x) = \sqrt{x-4}$, $a = 4$, $b = 8$, $c = 5$

COMPUTER EXPLORATIONS

In Exercises 91–94, you will find the area between curves in the plane when you cannot find their points of intersection using simple algebra. Use a CAS to perform the following steps:

- Plot the curves together to see what they look like and how many points of intersection they have.
 - Use the numerical equation solver in your CAS to find all the points of intersection.
 - Integrate $|f(x) - g(x)|$ over consecutive pairs of intersection values.
 - Sum together the integrals found in part (c).
- $f(x) = \frac{x^3}{3} - \frac{x^2}{2} - 2x + \frac{1}{3}$, $g(x) = x - 1$
 - $f(x) = \frac{x^4}{2} - 3x^3 + 10$, $g(x) = 8 - 12x$
 - $f(x) = x + \sin(2x)$, $g(x) = x^3$
 - $f(x) = x^2 \cos x$, $g(x) = x^3 - x$

Chapter 5

Questions to Guide Your Review

- How can you sometimes estimate quantities like distance traveled, area, and average value with finite sums? Why might you want to do so?
- What is sigma notation? What advantage does it offer? Give examples.
- What is a Riemann sum? Why might you want to consider such a sum?
- What is the norm of a partition of a closed interval?
- What is the definite integral of a function f over a closed interval $[a, b]$? When can you be sure it exists?
- What is the relation between definite integrals and area? Describe some other interpretations of definite integrals.
- What is the average value of an integrable function over a closed interval? Must the function assume its average value? Explain.
- Describe the rules for working with definite integrals (Table 5.4). Give examples.
- What is the Fundamental Theorem of Calculus? Why is it so important? Illustrate each part of the theorem with an example.
- What is the Net Change Theorem? What does it say about the integral of velocity? The integral of marginal cost?

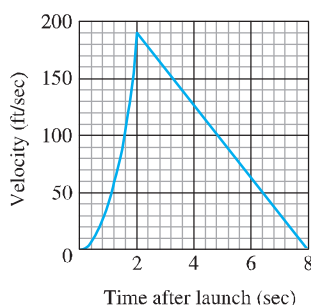
11. Discuss how the processes of integration and differentiation can be considered as “inverses” of each other.
12. How does the Fundamental Theorem provide a solution to the initial value problem $dy/dx = f(x)$, $y(x_0) = y_0$, when f is continuous?
13. How is integration by substitution related to the Chain Rule?
14. How can you sometimes evaluate indefinite integrals by substitution? Give examples.
15. How does the method of substitution work for definite integrals? Give examples.
16. How do you define and calculate the area of the region between the graphs of two continuous functions? Give an example.

Chapter 5

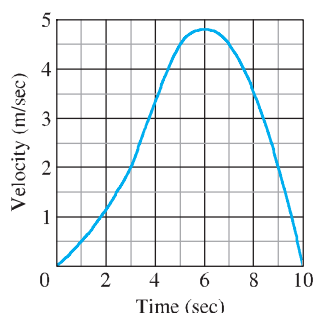
Practice Exercises

Finite Sums and Estimates

1. The accompanying figure shows the graph of the velocity (ft/sec) of a model rocket for the first 8 sec after launch. The rocket accelerated straight up for the first 2 sec and then coasted to reach its maximum height at $t = 8$ sec.



- a. Assuming that the rocket was launched from ground level, about how high did it go? (This is the rocket in Section 3.3, Exercise 17, but you do not need to do Exercise 17 to do the exercise here.)
- b. Sketch a graph of the rocket's height above ground as a function of time for $0 \leq t \leq 8$.
2. a. The accompanying figure shows the velocity (m/sec) of a body moving along the s -axis during the time interval from $t = 0$ to $t = 10$ sec. About how far did the body travel during those 10 sec?
- b. Sketch a graph of s as a function of t for $0 \leq t \leq 10$ assuming $s(0) = 0$.



3. Suppose that $\sum_{k=1}^{10} a_k = -2$ and $\sum_{k=1}^{10} b_k = 25$. Find the value of

- a. $\sum_{k=1}^{10} \frac{a_k}{4}$
- b. $\sum_{k=1}^{10} (b_k - 3a_k)$

- c. $\sum_{k=1}^{10} (a_k + b_k - 1)$
- d. $\sum_{k=1}^{10} \left(\frac{5}{2} - b_k \right)$

4. Suppose that $\sum_{k=1}^{20} a_k = 0$ and $\sum_{k=1}^{20} b_k = 7$. Find the values of

- a. $\sum_{k=1}^{20} 3a_k$
- b. $\sum_{k=1}^{20} (a_k + b_k)$
- c. $\sum_{k=1}^{20} \left(\frac{1}{2} - \frac{2b_k}{7} \right)$
- d. $\sum_{k=1}^{20} (a_k - 2)$

Definite Integrals

In Exercises 5–8, express each limit as a definite integral. Then evaluate the integral to find the value of the limit. In each case, P is a partition of the given interval and the numbers c_k are chosen from the subintervals of P .

5. $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n (2c_k - 1)^{-1/2} \Delta x_k$, where P is a partition of $[1, 5]$
6. $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n c_k (c_k^2 - 1)^{1/3} \Delta x_k$, where P is a partition of $[1, 3]$
7. $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n \left(\cos \left(\frac{c_k}{2} \right) \right) \Delta x_k$, where P is a partition of $[-\pi, 0]$
8. $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n (\sin c_k)(\cos c_k) \Delta x_k$, where P is a partition of $[0, \pi/2]$
9. If $\int_{-2}^2 3f(x) dx = 12$, $\int_{-2}^5 f(x) dx = 6$, and $\int_{-2}^5 g(x) dx = 2$, find the values of the following.

- a. $\int_{-2}^2 f(x) dx$
- b. $\int_2^5 f(x) dx$
- c. $\int_5^{-2} g(x) dx$
- d. $\int_{-2}^5 (-\pi g(x)) dx$
- e. $\int_{-2}^5 \left(\frac{f(x) + g(x)}{5} \right) dx$

10. If $\int_0^2 f(x) dx = \pi$, $\int_0^2 7g(x) dx = 7$, and $\int_0^1 g(x) dx = 2$, find the values of the following.

- a. $\int_0^2 g(x) dx$
- b. $\int_1^2 g(x) dx$
- c. $\int_2^0 f(x) dx$
- d. $\int_0^2 \sqrt{2} f(x) dx$
- e. $\int_0^2 (g(x) - 3f(x)) dx$

Area

In Exercises 11–14, find the total area of the region between the graph of f and the x -axis.

11. $f(x) = x^2 - 4x + 3, \quad 0 \leq x \leq 3$

12. $f(x) = 1 - (x^2/4), \quad -2 \leq x \leq 3$

13. $f(x) = 5 - 5x^{2/3}, \quad -1 \leq x \leq 8$

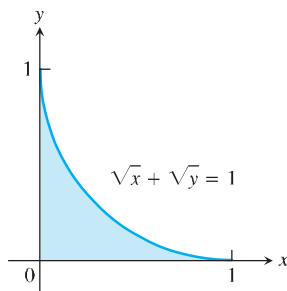
14. $f(x) = 1 - \sqrt{x}, \quad 0 \leq x \leq 4$

Find the areas of the regions enclosed by the curves and lines in Exercises 15–26.

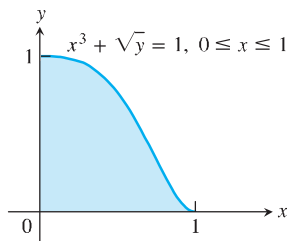
15. $y = x, \quad y = 1/x^2, \quad x = 2$

16. $y = x, \quad y = 1/\sqrt{x}, \quad x = 2$

17. $\sqrt{x} + \sqrt{y} = 1, \quad x = 0, \quad y = 0$



18. $x^3 + \sqrt{y} = 1, \quad x = 0, \quad y = 0, \quad \text{for } 0 \leq x \leq 1$



19. $x = 2y^2, \quad x = 0, \quad y = 3$ 20. $x = 4 - y^2, \quad x = 0$

21. $y^2 = 4x, \quad y = 4x - 2$

22. $y^2 = 4x + 4, \quad y = 4x - 16$

23. $y = \sin x, \quad y = x, \quad 0 \leq x \leq \pi/4$

24. $y = |\sin x|, \quad y = 1, \quad -\pi/2 \leq x \leq \pi/2$

25. $y = 2 \sin x, \quad y = \sin 2x, \quad 0 \leq x \leq \pi$

26. $y = 8 \cos x, \quad y = \sec^2 x, \quad -\pi/3 \leq x \leq \pi/3$

27. Find the area of the “triangular” region bounded on the left by $x + y = 2$, on the right by $y = x^2$, and above by $y = 2$.

28. Find the area of the “triangular” region bounded on the left by $y = \sqrt{x}$, on the right by $y = 6 - x$, and below by $y = 1$.

29. Find the extreme values of $f(x) = x^3 - 3x^2$ and find the area of the region enclosed by the graph of f and the x -axis.

30. Find the area of the region cut from the first quadrant by the curve $x^{1/2} + y^{1/2} = a^{1/2}$.

31. Find the total area of the region enclosed by the curve $x = y^{2/3}$ and the lines $x = y$ and $y = -1$.

32. Find the total area of the region between the curves $y = \sin x$ and $y = \cos x$ for $0 \leq x \leq 3\pi/2$.

Initial Value Problems

33. Show that $y = x^2 + \int_1^x \frac{1}{t} dt$ solves the initial value problem

$$\frac{d^2y}{dx^2} = 2 - \frac{1}{x^2}; \quad y'(1) = 3, \quad y(1) = 1.$$

34. Show that $y = \int_0^x (1 + 2\sqrt{\sec t}) dt$ solves the initial value problem

$$\frac{d^2y}{dx^2} = \sqrt{\sec x} \tan x; \quad y'(0) = 3, \quad y(0) = 0.$$

Express the solutions of the initial value problems in Exercises 35 and 36 in terms of integrals.

35. $\frac{dy}{dx} = \frac{\sin x}{x}, \quad y(5) = -3$

36. $\frac{dy}{dx} = \sqrt{2 - \sin^2 x}, \quad y(-1) = 2$

Evaluating Indefinite Integrals

Evaluate the integrals in Exercises 37–44.

37. $\int 2(\cos x)^{-1/2} \sin x \, dx$ 38. $\int (\tan x)^{-3/2} \sec^2 x \, dx$

39. $\int (2\theta + 1 + 2 \cos(2\theta + 1)) \, d\theta$

40. $\int \left(\frac{1}{\sqrt{2\theta - \pi}} + 2 \sec^2(2\theta - \pi) \right) d\theta$

41. $\int \left(t - \frac{2}{t} \right) \left(t + \frac{2}{t} \right) dt$ 42. $\int \frac{(t+1)^2 - 1}{t^4} dt$

43. $\int \sqrt{t} \sin(2t^{3/2}) \, dt$ 44. $\int \sec \theta \tan \theta \sqrt{1 + \sec \theta} \, d\theta$

Evaluating Definite Integrals

Evaluate the integrals in Exercises 45–70.

45. $\int_{-1}^1 (3x^2 - 4x + 7) \, dx$ 46. $\int_0^1 (8s^3 - 12s^2 + 5) \, ds$

47. $\int_1^2 \frac{4}{v^2} \, dv$

48. $\int_1^{27} x^{-4/3} \, dx$

49. $\int_1^4 \frac{dt}{t\sqrt{t}}$

50. $\int_1^4 \frac{(1 + \sqrt{u})^{1/2}}{\sqrt{u}} \, du$

51. $\int_0^1 \frac{36 \, dx}{(2x + 1)^3}$

52. $\int_0^1 \frac{dr}{\sqrt[3]{(7 - 5r)^2}}$

53. $\int_{1/8}^1 x^{-1/3} (1 - x^{2/3})^{3/2} \, dx$

54. $\int_0^{1/2} x^3 (1 + 9x^4)^{-3/2} \, dx$

55. $\int_0^\pi \sin^2 5r \, dr$

56. $\int_0^{\pi/4} \cos^2 \left(4t - \frac{\pi}{4} \right) \, dt$

57. $\int_0^{\pi/3} \sec^2 \theta \, d\theta$

58. $\int_{\pi/4}^{3\pi/4} \csc^2 x \, dx$

59. $\int_\pi^{3\pi} \cot^2 \frac{x}{6} \, dx$

60. $\int_0^\pi \tan^2 \frac{\theta}{3} \, d\theta$

61. $\int_{-\pi/3}^0 \sec x \tan x \, dx$

62. $\int_{\pi/4}^{3\pi/4} \csc z \cot z \, dz$

$$\begin{array}{ll}
 63. \int_0^{\pi/2} 5(\sin x)^{3/2} \cos x \, dx & 64. \int_{-1}^1 2x \sin(1 - x^2) \, dx \\
 65. \int_{-\pi/2}^{\pi/2} 15 \sin^4 3x \cos 3x \, dx & 66. \int_0^{2\pi/3} \cos^{-4}\left(\frac{x}{2}\right) \sin\left(\frac{x}{2}\right) \, dx \\
 67. \int_0^{\pi/2} \frac{3 \sin x \cos x}{\sqrt{1 + 3 \sin^2 x}} \, dx & 68. \int_0^{\pi/4} \frac{\sec^2 x}{(1 + 7 \tan x)^{2/3}} \, dx \\
 69. \int_0^{\pi/3} \frac{\tan \theta}{\sqrt{2 \sec \theta}} \, d\theta & 70. \int_{\pi^2/36}^{\pi^2/4} \frac{\cos \sqrt{t}}{\sqrt{t} \sin \sqrt{t}} \, dt
 \end{array}$$

Average Values

71. Find the average value of $f(x) = mx + b$
- over $[-1, 1]$
 - over $[-k, k]$
72. Find the average value of
- $y = \sqrt{3x}$ over $[0, 3]$
 - $y = \sqrt{ax}$ over $[0, a]$
73. Let f be a function that is differentiable on $[a, b]$. In Chapter 2 we defined the average rate of change of f over $[a, b]$ to be

$$\frac{f(b) - f(a)}{b - a}$$

and the instantaneous rate of change of f at x to be $f'(x)$. In this chapter we defined the average value of a function. For the new definition of average to be consistent with the old one, we should have

$$\frac{f(b) - f(a)}{b - a} = \text{average value of } f' \text{ on } [a, b].$$

Is this the case? Give reasons for your answer.

74. Is it true that the average value of an integrable function over an interval of length 2 is half the function's integral over the interval? Give reasons for your answer.

- T** 75. Compute the average value of the temperature function

$$f(x) = 37 \sin\left(\frac{2\pi}{365}(x - 101)\right) + 25$$

for a 365-day year. This is one way to estimate the annual mean air temperature in Fairbanks, Alaska. The National Weather Service's official figure, a numerical average of the daily normal mean air temperatures for the year, is 25.7°F , which is slightly higher than the average value of the approximating function $f(x)$.

- T** 76. **Specific heat of a gas** Specific heat C_v is the amount of heat required to raise the temperature of a given mass of gas with constant volume by 1°C , measured in units of cal/deg-mole (calories per degree gram molecule). The specific heat of oxygen depends on its temperature T and satisfies the formula

$$C_v = 8.27 + 10^{-5}(26T - 1.87T^2).$$

Find the average value of C_v for $20^\circ \leq T \leq 675^\circ\text{C}$ and the temperature at which it is attained.

Differentiating Integrals

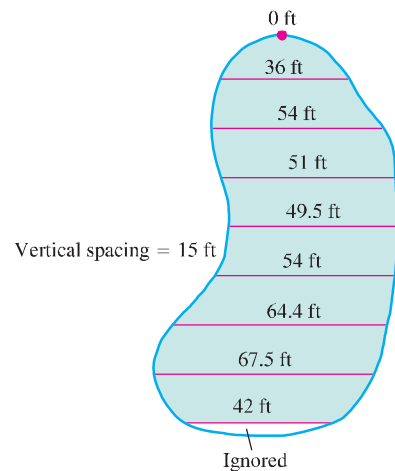
In Exercises 77–80, find dy/dx .

$$\begin{array}{ll}
 77. y = \int_2^x \sqrt{2 + \cos^3 t} \, dt & 78. y = \int_2^{7x^2} \sqrt{2 + \cos^3 t} \, dt
 \end{array}$$

$$\begin{array}{ll}
 79. y = \int_x^1 \frac{6}{3 + t^4} \, dt & 80. y = \int_{\sec x}^2 \frac{1}{t^2 + 1} \, dt
 \end{array}$$

Theory and Examples

81. Is it true that every function $y = f(x)$ that is differentiable on $[a, b]$ is itself the derivative of some function on $[a, b]$? Give reasons for your answer.
82. Suppose that $F(x)$ is an antiderivative of $f(x) = \sqrt{1 + x^4}$. Express $\int_0^1 \sqrt{1 + x^4} \, dx$ in terms of F and give a reason for your answer.
83. Find dy/dx if $y = \int_x^1 \sqrt{1 + t^2} \, dt$. Explain the main steps in your calculation.
84. Find dy/dx if $y = \int_{\cos x}^0 (1/(1 - t^2)) \, dt$. Explain the main steps in your calculation.
85. **A new parking lot** To meet the demand for parking, your town has allocated the area shown here. As the town engineer, you have been asked by the town council to find out if the lot can be built for \$10,000. The cost to clear the land will be \$0.10 a square foot, and the lot will cost \$2.00 a square foot to pave. Can the job be done for \$10,000? Use a lower sum estimate to see. (Answers may vary slightly, depending on the estimate used.)



86. Skydivers A and B are in a helicopter hovering at 6400 ft. Skydiver A jumps and descends for 4 sec before opening her parachute. The helicopter then climbs to 7000 ft and hovers there. Forty-five seconds after A leaves the aircraft, B jumps and descends for 13 sec before opening his parachute. Both skydivers descend at 16 ft/sec with parachutes open. Assume that the skydivers fall freely (no effective air resistance) before their parachutes open.
- At what altitude does A's parachute open?
 - At what altitude does B's parachute open?
 - Which skydiver lands first?